

LABWORK 5

This document contains 10 real-life themed programming questions focused on working with arrays and functions in C.

Function 1: find_max_temperature

- **Return Type:** `float`
 - **Parameters:** `float temps[7]`
 - **Description:** A city weather app records daily temperatures for a week. Write a function that returns the highest temperature recorded.
 - **Example:**
 - Input: {22.5, 24.0, 21.8, 25.3, 23.9, 26.1, 24.7}
 - Output: 26.1
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Function 2: array_copy_reverse

- **Return Type:** `void`
 - **Parameters:** `int source[], int dest[], int size`
 - **Description:** A software system stores daily visitor counts. Write a function that copies the data from one array to another in reverse order.
 - **Example:**
 - Input: source = {100, 200, 300, 400}
 - Output: dest = {400, 300, 200, 100}
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Function 3: find_second_minimum

- **Return Type:** `int` **Parameters:** `int arr[], int size`
 - **Description:** A company tracks weekly defect counts from production lines. Write a function that returns the second smallest number in the array.
 - **Example:**
 - Input: {12, 8, 15, 9, 20}
 - Output: 9
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Function 4: count_even_numbers

- **Return Type:** `int`
 - **Parameters:** `int arr[], int size`
 - **Description:** A sensor records integer readings throughout the day. Write a function that counts how many even numbers exist in the dataset.
 - **Example:**
 - Input: {3, 8, 5, 10, 12, 7}
 - Output: 3
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Function 5: replace_negative_values

- **Return Type:** `void`
 - **Parameters:** `int arr[], int size`
 - **Description:** A system logs financial transactions. Negative values represent errors. Write a function that replaces all negative numbers with 0.
 - **Example:**
 - Input: {120, -40, 300, -15, 90}
 - Output: {120, 0, 300, 0, 90}
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Function 6: shift_right

- **Return Type:** `void`
 - **Parameters:** `int arr[], int size`
 - **Description:** Write a function that performs a one-position right shift on an integer array (the last element moves to the first position).
 - **Example:**
 - Input: {1, 2, 3, 4, 5}
 - Output: {5, 1, 2, 3, 4}
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Function 7: merge_two_arrays

- **Return Type:** `void`
 - **Parameters:** `int arr1[], int arr2[], int merged[], int size1, int size2`
 - **Description:** Two delivery trucks record their package counts daily. Write a function that merges both arrays into a single array.
 - **Example:**
 - Input: `arr1 = {5, 7, 9}, arr2 = {4, 6, 8, 10}`
 - Output: `merged = {5, 7, 9, 4, 6, 8, 10}`
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Function 8: search_in_array

- **Return Type:** `int`
 - **Parameters:** `int arr[], int size, int value`
 - **Description:** A library stores book IDs in an array. Write a function that searches for a given book ID and returns its index. Return -1 if not found.
 - **Example:**
 - Input: {101, 204, 356, 478}, value = 356
 - Output: 2
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Function 9: compare_sales_data

- **Return Type:** `int`
- **Parameters:** `float branch1[], float branch2[], int size`
- **Description:** Two store branches record their daily sales. Write a function that compares them and returns the number of days where Branch 1 > Branch 2.

- **Example:**
 - Input: branch1 = {200.5, 180.0, 250.0}, branch2 = {190.0, 200.0, 240.0}
 - Output: 2
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Function 10: normalize_data

- **Return Type:** `void`
 - **Parameters:** `float data[], int size`
 - **Description:** Normalize all numbers in an array by dividing each element by the maximum value. (Each result becomes a value between 0 and 1.)
 - **Example:**
 - Input: {10.0, 20.0, 30.0, 40.0}
 - Output: {0.25, 0.5, 0.75, 1.0}
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Main Function (15 pts)

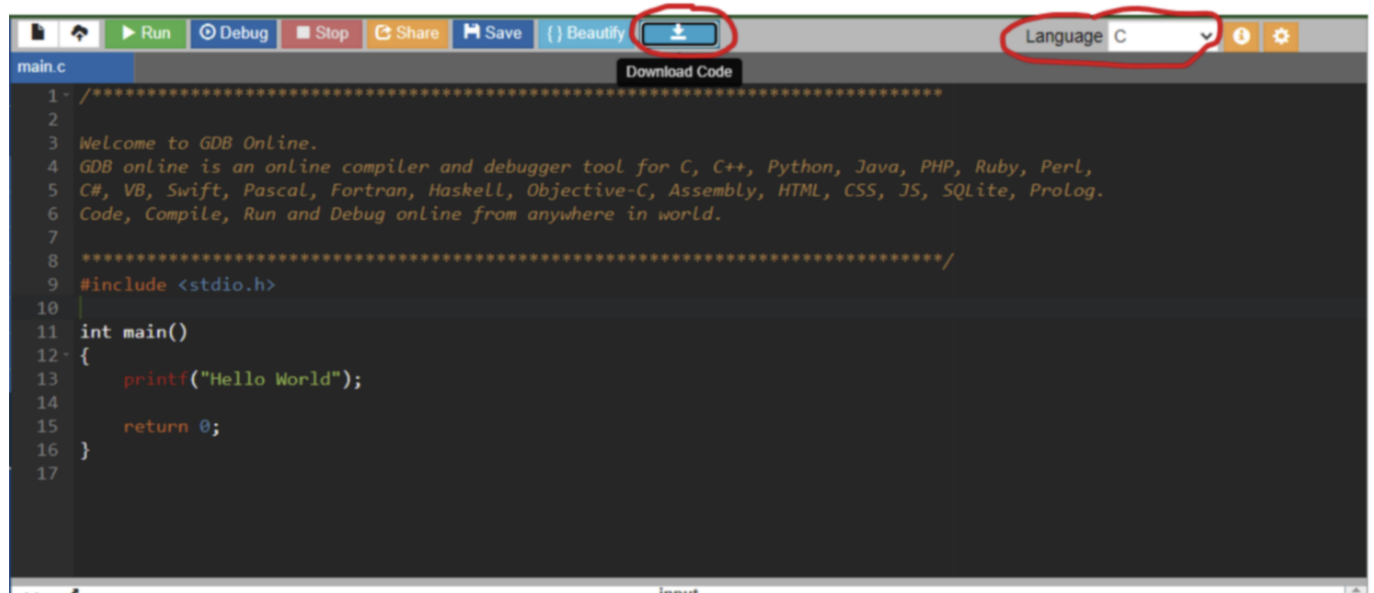
- **Function Name:** `main`
 - **Return Type:** `int`
 - **Parameters:** `none`
 - **Description:**
 - Ask the user to enter input values for arrays and call all the above functions to demonstrate:
 - Print results clearly and formatted.
 - Demonstrate function calling, passing arrays, and pointer behavior.
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Submission Format

- Submit your `.c` file named as:
`name_surname.c`
- Late submissions will not be accepted.
- Follow function naming and signature exactly.
- Clear screen and print statements are optional, not graded.

Restrictions

- If you are not sure something is free to use or not, please ask your assistant.
- You have to complete the labwork using the functions provided. If you complete the labwork without using functions, your work will not be graded.
- Mobile phone and internet usage are not allowed.
- You can only access yulearn and the online C compiler <https://www.onlinegdb.com/> Do not forget to select the language if you use onlinegdb. When you finish your work, you can download your code by using the download code button on top of the window.



```
1  /*****
2
3  Welcome to GDB Online.
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,
5  C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.
6  Code, Compile, Run and Debug online from anywhere in world.
7
8  *****/
9  #include <stdio.h>
10
11 int main()
12 {
13     printf("Hello World");
14
15     return 0;
16 }
17
```